

| Mapped CO(s) Number | Practical/ Lab Session Outcomes (LSOs)                                                                                                                                                                                | Laboratory Experiment/ Practical Titles                                                                                                   |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| CO1                 | <i>LSO 1.1</i> Use manual drawing instruments and BIS conventions properly.<br><i>LSO 1.2</i> Identify various types of lines and uses in engineering drawing                                                         | LE- 1 Basics of Engineering Drawing.                                                                                                      |
| CO1                 | <i>LSO 2.1</i> Draw engineering curves and engineering scales accurately using appropriate methods.                                                                                                                   | LE- 2 Geometrical constructions of lines, circles, curves, tangents and regular polygons.                                                 |
| CO2                 | <i>LSO 3.1</i> Draw projections of points and lines.<br><i>LSO 3.2</i> Analyze the concepts of Orthographic Projections                                                                                               | LE- 3 Orthographic Projection First/Third Angle method of projections, projection of points & projection of lines in different positions. |
| CO2, CO3            | <i>LSO 4.1</i> Drawing 2D Views using Orthographic Projections<br><i>LSO 4.2</i> Apply orthographic projection principles to obtain true length and true shape in given situation.                                    | LE- 4 Drawing views of objects (plane and solid) using Orthographic Method of Projection.                                                 |
| CO3                 | <i>LSO 5.1</i> Draw projections of solids in different orientations.<br><i>LSO 5.2</i> Analyze the concepts of Isometric Projections<br><i>LSO 5.3</i> Draw sectional views and determine the true shape of sections. | LE- 5 Orthographic Projection of solids (prisms, pyramids, cylinders, cones and auxiliary views) and also sectional view.                 |
| CO3                 | <i>LSO 6.1</i> Draw multiview orthographic projections of engineering objects.<br><i>LSO 6.2</i> Interpret pictorial and orthographic views.                                                                          | LE- 6 Conversion between pictorial and orthographic views.                                                                                |
| CO4                 | <i>LSO 7.1</i> Drawing 3D objects on 2D surfaces using Isometric Projections.                                                                                                                                         | LE- 7 Drawing objects using Isometric Projections.                                                                                        |
| CO4                 | <i>LSO 8.1</i> Convert orthographic views into isometric views.                                                                                                                                                       | LE- 8 Orthographic–isometric conversion.                                                                                                  |
| CO4                 | <i>LSO 9.1</i> Develop lateral surfaces of solids                                                                                                                                                                     | LE- 9 Drawing objects using Development of surfaces (prisms, pyramids, cylinders and cones)                                               |
| CO5                 | <i>LSO 10.1</i> Use commands in draw, modify, annotate and layer panels                                                                                                                                               | LE- 10 Introduction to AutoCAD – 2D drawing.                                                                                              |
| CO5                 | <i>LSO 11.1</i> Develop 2D drawings using CAD software for orthographic projection                                                                                                                                    | LE- 11 Develop 2D drawings using CAD software for a given engineering drawing – drawings of like screw, nut, bolt.                        |

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| CO5                 | <i>LSO 12.1</i> Develop 2D drawings using CAD software for an isometric project.     | LE- 12 Develop 2D drawings using CAD software for a given engineering drawing – drawing of engineering components. |
| CO5                 | <i>LSO 13.1</i> Prepare 2D engineering drawings using CAD                            | LE- 13 Preparation of orthographic, sectional and isometric drawings of engineering components/layouts.            |
| CO5                 | <i>LSO 14.1</i> Prepare 2D drawings in CAD software for physical engineering objects | LE- 14 Preparation of basic 3D models and engineering drawings of machine components,                              |